NAEEM NAIRO

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SUMMARY

Biomedical and mechatronics engineer with 3+ years designing assistive mobility devices from concept to mass production. Certified SOLIDWORKS Expert with strong expertise in CAD, FEA, DFM, GD&T, prototyping, and verification testing, plus sensor integration and embedded systems (ESP32, STM32). Focused on translating clinical and user needs into functional devices user can rely on every day.

ENGINEERING EXPERIENCE

Product Design Engineer, Full Time | AHL Logics

June 2022-August 2025

- Led the development of assistive mobility devices (strollers, walkers, baby car seats), from the conceptual stage Through prototyping and testing, ending with mass production and product launching.
- Designed and optimized assemblies and components in SOLIDWORKS, applying DFM principles to improve manufacturability while reducing cost and part complexity.
- Conducted finite element analysis (FEA) to improve structural integrity and durability while reducing weight, then validated predictions against physical prototype testing.
- Produced design specifications, test plans/reports, technical drawings, BOMs, assembly instructions and manufacturing documentation.
- Coordinated with manufacturing, quality, and suppliers throughout design transfer, aligning drawings and BOMs while resolving DFM and tooling issues ahead of production.

Product Design Engineer, Part Time | AHL Logics

November 2021-June 2022

- Supported concept refinement by collaborated on mechanism design, material selection, prototype builds.
- Supported design reviews, prototype evaluation, FMEA, root cause analysis, verification/validation and engineering changes while collaborating with customers, suppliers and manufacturers.

Teaching Assistant, Instrumentation Laboratory | University of New Haven

New Haven, CT | Fall 2025-Present

- Serve as Teaching Assistant for the Instrumentation and Sensors Laboratory, guiding students through LabVIEW and NI ELVIS experiments in EMG signal acquisition and temperature sensing.
- Adapt a ball and beam control platform into a rehabilitation tool, developing training strategies for patients with post-stroke mobility impairments.

EDUCATION

M.S. Biomedical Engineering | University of New Haven, Tagliatela College of Engineering

Relevant coursework: Biomedical Polymers; Biomedical Sensors and Instrumentation; Biostatistics; Biomechanics; Signal analysis.

B.S. Mechanical Engineering (Mechatronics) | PPU | January 2022

Relevant coursework: Mechanical design; Material science; Control system; Manufacturing processes; Mechatronic systems; Thermodynamic

TECHNICAL SKILLS

Quality & Regulatory: design verification and validation (V&V), FMEA, root-cause analysis, ISO 13485 and FDA 21 CFR Part 820.
Medical & Assistive Technology: prosthetics, rehabilitation devices, mobility products, biomedical sensors, human-centered design.
Analysis & Simulation: SOLIDWORKS Simulation, ANSYS, structural/static FEA, MATLAB, prototype testing and validation
Instrumentation & Embedded Systems: ESP32, STM32, I2C, LabVIEW, NI ELVIS, Arduino.
Design & Prototyping: SOLIDWORKS (CSWE), CAD modeling, DFM, GD&T.

CERTIFICATIONS

- Certified SOLIDWORKS Expert (CSWE), December 2025
- Certified SOLIDWORKS Professional Simulation, February 2025
- SOLIDWORKS Additive Manufacturing Associate, January 2025
- Additive Manufacturing: Materials for 3D Printing, December 2025

LEADERSHIP / AWARDS

- Fulbright Scholar, U.S. government international scholarship recipient, 2025-2027.
- Secretary, Biomedical Engineering Society, University of New Haven, Spring 2026-Present; support events, professional coordination and member communications.
- President, Mechanical Engineering Society, Palestine Polytechnic University, Spring 2021.
- Professional affiliations: IEEE, ASME and Biomedical Engineering Society (BMES)
- SOLIDWORKS Champion, recognized for advanced proficiency in 3D CAD modeling, and product development.